

 AMERICAN UNIVERSITY OF PHNOM PENH STUDY LOCALLY. LIVE GLOBALLY.	American University of Phnom Penh		
	School of Digital Technologies		
	ITM-390: Machine Learning		
Research Proposal			

Project Information			
Project title	English	Khmer Transliteration Keyboard	
	Khmer	ប្រព័ន្ធវាយអក្សរខ្មែរតាមអក្សរឡាតាំង	
Advisor	kuntha pin	Department	ITM - Machine learning - 390
Team Members			
Team Leader	Kimyoo, Ngov	Student ID	2024523
Member 1	Elsewin, Lim	Student ID	2024215
Member 2	Kimleng, Sok	Student ID	2025224
Member 3	Suos monychanrotha	Student ID	2025112

I. Introduction

This project focuses on developing a Romanized Khmer-to-Khmer script input system that converts Khmer words typed using Latin letters into proper Khmer script. In daily digital communication, many Khmer speakers informally type Khmer sounds using English letters, such as “kom” for “កំ” or “nh” for “ង”^១. However, this type of Romanized Khmer does not follow one fixed spelling standard, which creates ambiguity because one Romanized input may correspond to several possible Khmer words.

Existing studies have explored Khmer predictive text entry and Romanized Khmer transliteration, showing that Khmer input is challenging due to the structure of the Khmer writing system, including consonant classes, dependent vowels, subscripts, and diacritics.

This project proposes a hybrid Natural Language Processing approach for predicting and ranking Khmer word candidates from Romanized input. The system will use a dataset of Romanized Khmer and Khmer-script word pairs, such as “kom” → “កំ”, together with candidate-generation rules based on Khmer spelling patterns, such as “k” → “ក គ ឃ”, “kh” → “ខ ឃ”. A machine learning model will then rank the most likely Khmer outputs using features such as spelling similarity, word frequency, previous-word context, and user selection history.

The main hypothesis of this study is that combining rule-based candidate generation with machine learning-based ranking will produce more accurate and user-friendly predictions than a fixed dictionary lookup alone. The expected result of this project is to connect with keyboard and display an UI as user typed that can suggest ranked Khmer-script candidates from Romanized typing, helping users type Khmer more efficiently

II. Research Goals and Objectives

- a) **Main goal:** The main goal of this project is to make Khmer typing faster and more convenient. Adapting Romanized Khmer also helps create a standardized typing system that can be typed based on word sound alone.
- b) **Specific objectives:**
- To collect and prepare a dataset of Romanized Khmer and Khmer-script word pairs, including common words, informal spellings, and more complex Khmer words.
 - To create customized character and syllable mapping rules that connect Romanized input patterns with possible Khmer characters, vowels, subscripts, and diacritics.
 - To assign frequency scores to commonly used Khmer words, especially words frequently typed in Romanized form by young people in online communication.
 - To search and match user input with existing Romanized Khmer words in the dataset.
 - To generate additional Khmer word candidates using character-level and syllable-level rules when the exact input is not found in the dataset.
 - To score and rank candidate Khmer words based on spelling similarity, word frequency, previous-word context, and user selection history.
 - To save the user's selected word in order to improve future prediction accuracy and personalize the ranking results over time.

III. Literature review

Romanized Khmer-to-Khmer script conversion is related to Natural Language Processing (NLP), machine transliteration, and predictive text input. Transliteration refers to converting text from one writing system into another while preserving pronunciation or intended meaning. In the context of this study, the task is backward transliteration, where users type Khmer words using Latin letters, and the system predicts the most suitable Khmer-script output. This is challenging because Romanized Khmer is commonly used in informal digital communication, but it does not follow one fixed spelling standard. For example, the Khmer word “” may be typed as “khnhom,” “knhom,” “knyom,” or other variations such as short cut forms “nhom” or “nh” that preferred by most among young people. Therefore, a simple rule-based mapping is not enough to accurately identify the user’s intended word.

More recently, Nhem [\[1\]](#) proposed AkaraAlpha, a deep learning model for Romanized Khmer-to-Khmer script transliteration. The study employed a character-level sequence-to-sequence architecture with a Bidirectional Gated Recurrent Unit (BiGRU) encoder and an attention mechanism to convert informal Romanized Khmer text into standard Khmer script. While the model demonstrated promising transliteration performance, its effectiveness may be constrained by the wide variation in Romanization practices among Khmer speakers. Different age groups and communities often represent the same Khmer sounds using different Romanized spellings, resulting in inconsistencies that may not be fully captured by a fixed deep learning model. This limitation is particularly significant because younger users frequently adopt abbreviated or informal spellings that differ from the patterns observed in the training data [\[1\]](#).

The complexity of the Khmer writing system presents additional challenges for transliteration. According to the Library of Congress Khmer Romanization Table [\[2\]](#), Khmer script consists of consonants, dependent vowels, diacritical marks, and subscript consonant forms that interact to represent pronunciation. Furthermore, Khmer text is commonly written without spaces separating individual words, making word boundary identification more difficult. These linguistic characteristics complicate direct character-level transliteration because a single Romanized input may correspond to

multiple valid Khmer spellings. As a result, transliteration systems must address both orthographic ambiguity and contextual interpretation to generate accurate outputs [2].

Another challenge arises from the existence of multiple Romanization conventions for Khmer. Khmer sounds have been represented using various Romanization systems developed for different purposes, including academic, governmental, and informal online communication. Consequently, a single Khmer word may have several acceptable Romanized forms, increasing ambiguity during transliteration. This variation highlights the need for transliteration systems that can accommodate diverse spelling patterns rather than relying solely on fixed one-to-one character mappings. Such flexibility is essential for improving transliteration accuracy in real-world applications where users may follow different Romanization conventions [3].

3.1. Limitations

The transliteration dataset, developed by Chhunneng(n.d), provides a valuable collection of approximately 1,000 common Khmer words for each file (10 files in total with 10,000 total words) and their corresponding Romanized forms, serving as a useful resource for Khmer language processing tasks. However, the dataset primarily functions as a static dictionary and does not incorporate mechanisms for handling out-of-vocabulary (OOV) words or variations in Romanized spelling (Chhunneng, n.d.). Furthermore, the approach does not consider contextual information, word frequency, or user preferences when generating transliteration outputs. As a result, the system may encounter difficulties when processing user inputs that deviate from the predefined transliteration patterns or when multiple Khmer words share similar Romanized representations.

In addition, the Romanized Khmer-to-Khmer transliteration work by Netra AI Lab, focuses mainly on data preprocessing and the preparation of transliteration pairs for machine learning applications (Netra AI Lab, n.d.). While the project contributes significantly to the development of Khmer NLP resources, its primary emphasis is on dataset construction rather than end-user transliteration

functionality. Consequently, the system does not provide advanced candidate generation, contextual disambiguation, adaptive ranking, or personalized prediction capabilities. In situations where the user input contains spelling inconsistencies, uncommon Romanization patterns, or ambiguous transliterations, the preprocessing-oriented approach may be insufficient to identify the most appropriate Khmer output.

To address these limitations, the proposed study introduces a hybrid Romanized Khmer-to-Khmer transliteration framework that combines dictionary-based retrieval with rule-based candidate generation. When a Romanized input is found in the dataset, the system retrieves possible Khmer candidates; otherwise, additional candidates are generated through customized character and syllable mapping rules. The generated candidates are then evaluated using multiple scoring features, including spelling similarity, word frequency, previous-word context, and user selection history. This hybrid approach extends beyond the capabilities of previous studies by improving the handling of unseen inputs, reducing transliteration ambiguity, and providing adaptive predictive text functionality. Consequently, the proposed system aims to achieve more accurate and context-aware transliteration while enhancing the overall user experience.

IV. Methodology

This study proposes a hybrid Romanized Khmer-to-Khmer script prediction system. The system combines dataset-based word matching, customized character and syllable mapping rules, and machine learning-based candidate ranking.

The project follows a hybrid approach. First, it searches the dataset for exact or similar Romanized Khmer inputs. If the input is found, the system retrieves possible Khmer candidates. If the input is not found, the system generates additional candidates using customized character and syllable mapping rules as a hybrid approach. After that, the system scores and ranks all candidates based on multiple features, including spelling similarity, word frequency, previous-word context, and user selection history.

This hybrid approach is selected because Romanized Khmer does not follow a single spelling standard. A purely rule-based system may fail when users type informal or shortened words, while a fixed dictionary system may fail when the input does not exist in the dataset. By combining both methods, the system can provide more flexible and accurate predictions.

This project will use two existing sources as the foundation of the dataset.

- <https://github.com/Chhunneng/khmer-text-transliteration/tree/main/1000-most-common-khmer-words>
- https://github.com/netra-ai-lab/Romanized-Khmer-to-Khmer-Transliteration/blob/main/data_preprocessing.py

We will customize mapping rules between Romanized input patterns and Khmer characters or syllables. For example, “k” may map to “ក”, “kh” may map to “ខ”, and “nh” may map to “ន”. However, the system will not depend only on direct character mapping because Khmer contains dependent vowels, subscripts, and diacritics. Therefore, the mapping rules will be used mainly to generate possible candidates when no exact dataset match is found.

The system process can be break down as follows:

1. Clean the input : handle user input such as remove extra spacing, case sensitivity, check spelling
2. Search the dataset : search matching Romanized word in the dataset
3. Generate possible Khmer candidates : Generate extra candidates based on character mapping rules
4. Score each candidate : rank candidate from most likely to least likely
5. Rank the best result first : display top suggestions to the user on the displayed UI
6. Save the user’s selected answer : when users select a suggested word, save it for future ranking improvement.

For candidate selection, we plan to use **RapidFuzz** library. **RapidFuzz** is a fast Python library for fuzzy string matching. It can compare strings and return a similarity score. For example: user input “khnhom” which is similar to possible output such as “knhom”, “knhum”, “nhom”, “nhum”.

V. Timeline of Plan

Break down your research into smaller tasks and assign a timeline for completion.

Task/	Week 1 (1–7)	Week 2 (8-14)	Week 3 (15-21)	Week 4 (22-28)
Plan	Initial idea phase. We began a small discussion.	Consulted the professor regarding this topic.		Assign roles among team members
Find/Train data			2 samples of dataset has been collected We began our research on the topic	
Coding				Setup the project, assign roles among team members, decide on techstacks, and begin coding.
Writing proposal				We collectively contribute to write the proposal all together.

VI. Expected Results

We expected this project to help the young generation type faster where digital communication is taking shape more than traditional handwriting. This study would strengthen the Romanized Khmer to teach a more standardized writing system. This is not limited to only young people, the older generation are welcome to learn to adapt to current changes of society behavior. Moreover, this will be a great inspiration for any future developer in future projects.

VII. Final Goal of the Research and Development Project

Indicate the long-term goal of your project:

- ☒ Intended for only ML Class
- ☐ Aiming for publication in a journal
- ☐ Planning to extend the research into a startup
- ☒ Other: Deepend our understanding on NLP

VIII. Expected effects of the development project

This project is expected to improve the Khmer text by developing a simple word prediction system that converts Romanized Khmer/sing Khmer (Khmer words but written in English) into proper Khmer words. This project aims to help people who are more familiar with English keyboards or people who find the Khmer Keyboard a bit more confusing. Additionally, the project will be at great aid to learners who struggle to understand the vowel sounds of the Khmer words.

The research contributes to the development of Khmer Natural Language Processing (NLP) technologies by providing methods for recognizing and predicting Khmer words from their Romanized forms. This can improve the accuracy and speed of Khmer text entry while reducing typing errors.

In real-world applications, the system could be integrated into mobile keyboards, messaging applications, educational platforms, and online communication tools. It can help users communicate more efficiently in the Khmer language and make Khmer digital content more accessible. Overall, the project supports the growth of Khmer language technology and encourages greater use of the Khmer language in digital environments

IX. Students' Role and Experience

Outline each student's specific role in the project and their relevant experience.

Student Names	Role in Project	Experience
Suos monychanrotha	Helping the team with project/proposal writing	Teamwork, writing a proposal, and research
Kimleng Sok	completing the proposal, and providing relevant ideas	research, code debugging and supporting
Kimyou Ngov	Team's leader, set up repository and coding environment, train the model	Leadership, bring ideas into reality, have worked with similar NLP concept project
Elsewin Lim	Team's coordinator, assist cleaning dataset, testing	Had involved and developed NLP projects and other ML projects

X. References

- [1] D. Nhem, "AkaraAlpha: An Efficient Romanized Khmer-Khmer Script Transliteration Model," Royal University of Phnom Penh, Apr. 2026.
https://www.researchgate.net/publication/403923747_AkaraAlpha_An_Efficient_Romanized_Khmer-Khmer_Script_Transliteration_Model
- [2] Library of Congress, "Khmer Romanization Table," ALA-LC Romanization Tables, 2013.
<https://www.loc.gov/catdir/cpsd/romanization/khmer.pdf>
- [3] Wikipedia contributors, "Khmer script," *Wikipedia, The Free Encyclopedia*.
https://en.wikipedia.org/wiki/Khmer_script

Rubric: Research Proposal Report (15 Points)

	Points	
Section	Moderate(half)	Good(full grade)
Introduction	1	2
Research Goals and Objectives	1	2
Literature Review	1	2
Methodology	1	2
Timeline of Plan	0.5	1
Expected Results	1	2
Expected effects of the development	0.5	1
Student's Rule	0.5	1
References	0.5	1
Correct format + human writing (not AI generate text).	0.5	1

Rubric: Creative Topic Idea (10 Points)

A topic can be considered creative if it satisfies **at least one** of the following aspects. The score depends on how strong the idea is.

No	Creativity Type	Description
1	New Application with Societal Impact	Applying AI/ML methods to a new real-world problem that can provide meaningful benefits to society.
2	Improvement to Existing Methods	Proposing a modification or improvement to an existing method to enhance performance such as classification accuracy, regression performance, computational speed, robustness, or efficiency.
3	Novel Algorithm or Method	Introducing a new algorithm or methodological innovation , including preprocessing, feature extraction, model architecture, or training strategy.
4	Novel Research Topic (Research Gap)	Investigating a new or underexplored research problem with limited previous studies.
5	New Dataset Design	Creating a new dataset that addresses limitations of existing datasets and supports solving an important real-world problem.

Simple Scoring Guide

Score	Interpretation
9-10	Very creative idea with strong novelty and clear research contribution
7 – 8	Creative idea but with moderate novelty
5 – 6	Some creativity but similar to existing work
3 – 4	Weak creativity; only small modification
0 – 2	No clear creative contribution